

PlaylistPro: Algorithmic Churn Audit

Auditing Fairness & Business Impact in Predictive Systems

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The Goal: Identifying At-Risk Subscribers

Our ADS predicts subscriber cancellation to allocate financial retention discounts fairly and effectively.

| Inputs & Outputs



Synthetic Data Input

125,000 users. Features include demographics (Age, Location) and behavioral metrics (Weekly Hours, Service Inquiries, Skip Rates).



System Output

Binary Class: 1 (Churned) or 0 (Retained). Thresholded at 0.5 Sigmoid probability to determine discount eligibility.

| Dataset Creator's Response

// The features that are correlated are based on prior business experience with SaaS companies, but not music streaming companies in particular. //

— Michael O'Donnell, Decision Sciences and Marketing Department, Adelphi University

Confirms a fundamental **domain mismatch**: the ADS treats music streaming like a B2B software trial.

Baseline MLP Performance

84%

Aggregate Accuracy

Kaggle Validation Metrics

While the model shows strong discriminative power (0.93 AUC-ROC), our audit proves these metrics mask severe demographic disparities.



Recall: 88%



F1-Score: 85%

Defining Fairness Benchmarks



Selection Rate

Applying the EEOC's 80% Rule to ensure equitable distribution of financial discounts across subgroups.



False Negative Rate

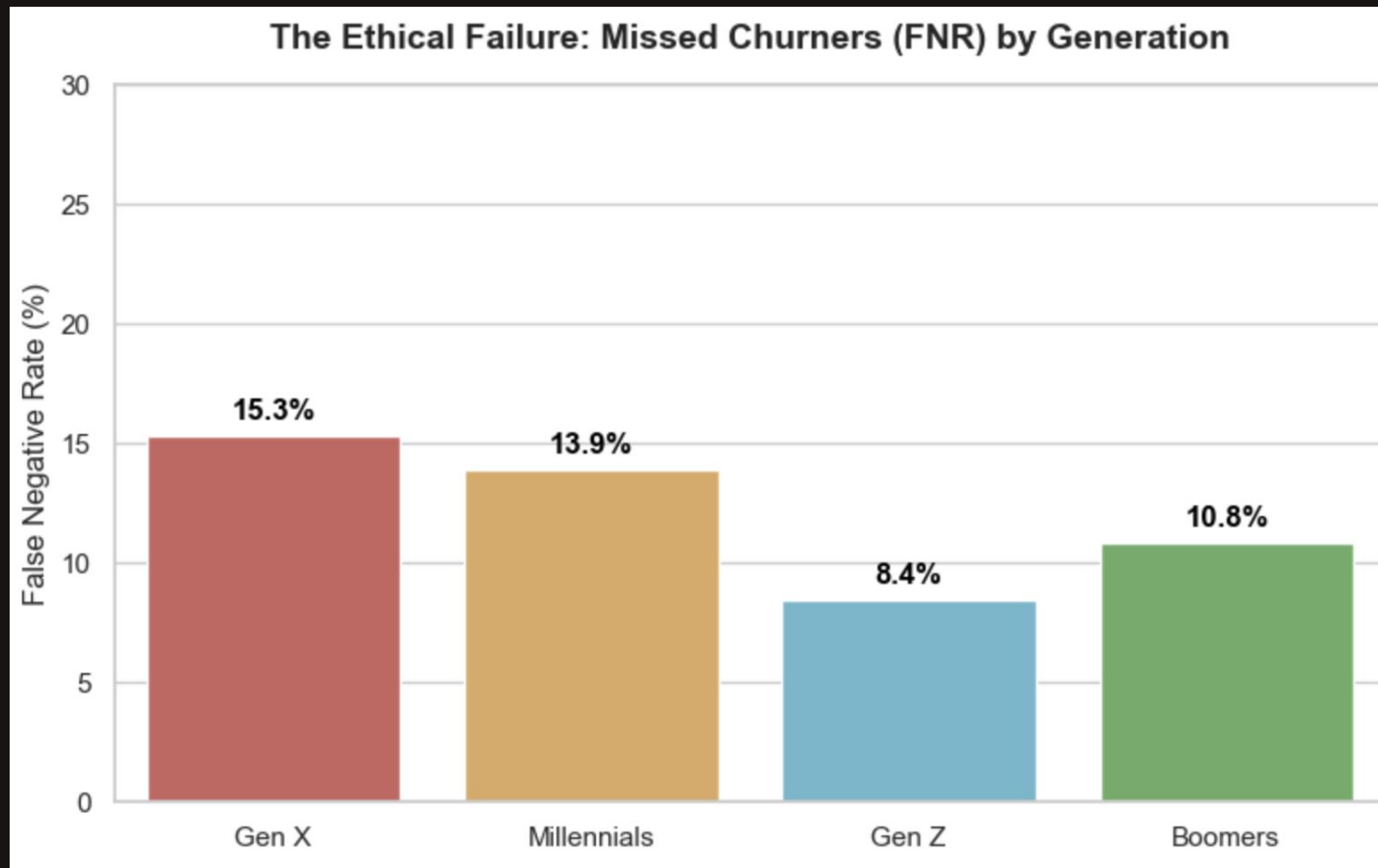
Focusing on "Missed Churners": users denied a benefit they deserved due to structural predictive error.



Dollar Impact

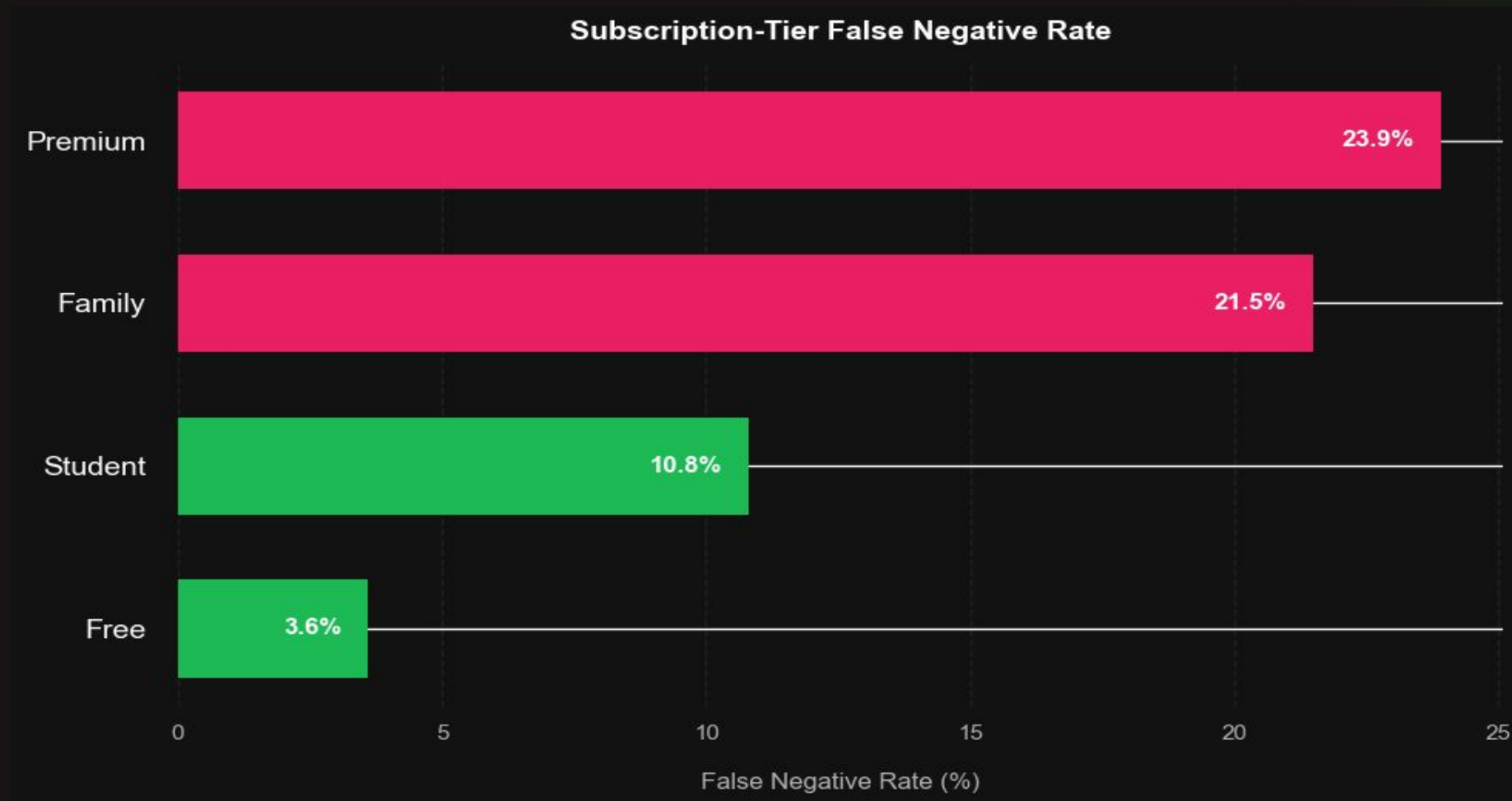
Translating error gaps into real-world marketing budget misallocation (\$65,200 total simulation).

Generational FNR Disparities



Ethical Failure: Gen X users are 1.8x more likely to be missed by the model compared to Gen Z.

The Subscription Tier Bias

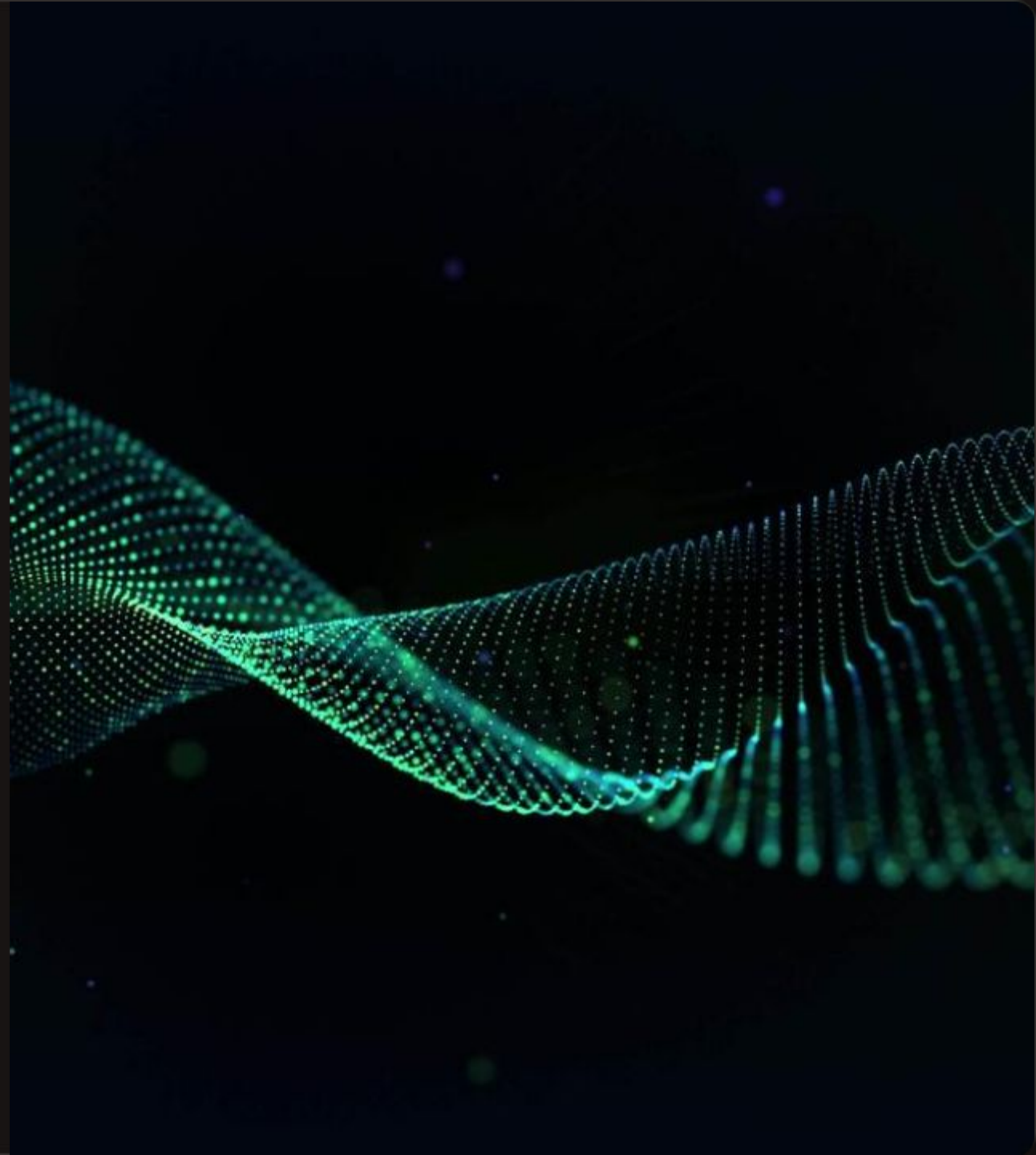


Business Paradox: The model is 6.6x more likely to miss churn signals from high-paying Premium customers.

Compounding Harm: Intersectionality

- ✓ Millennials - Premium: 30.4% FNR (Most Harmed)
- ✓ Gen X - Premium: 27.7% FNR
- ✓ Gen Z - Free: 1.7% FNR (Most Favored)

Being a paying adult in the core revenue demographic creates an ~18x disparity in predictive error.



| Interpretability: Behavioral Proxies

12.7%

CS Inquiries Impact

9.9%

Weekly Hours Impact

Permutation Importance reveals the model relies on behavioral proxies that penalize busy adults.

- ✓ CS Inquiries = Strongest Predictor
- ✓ Weekly Hours = Dominant Factor
- ✓ Subscription Tier = Model Shortcut

Financial Misallocation

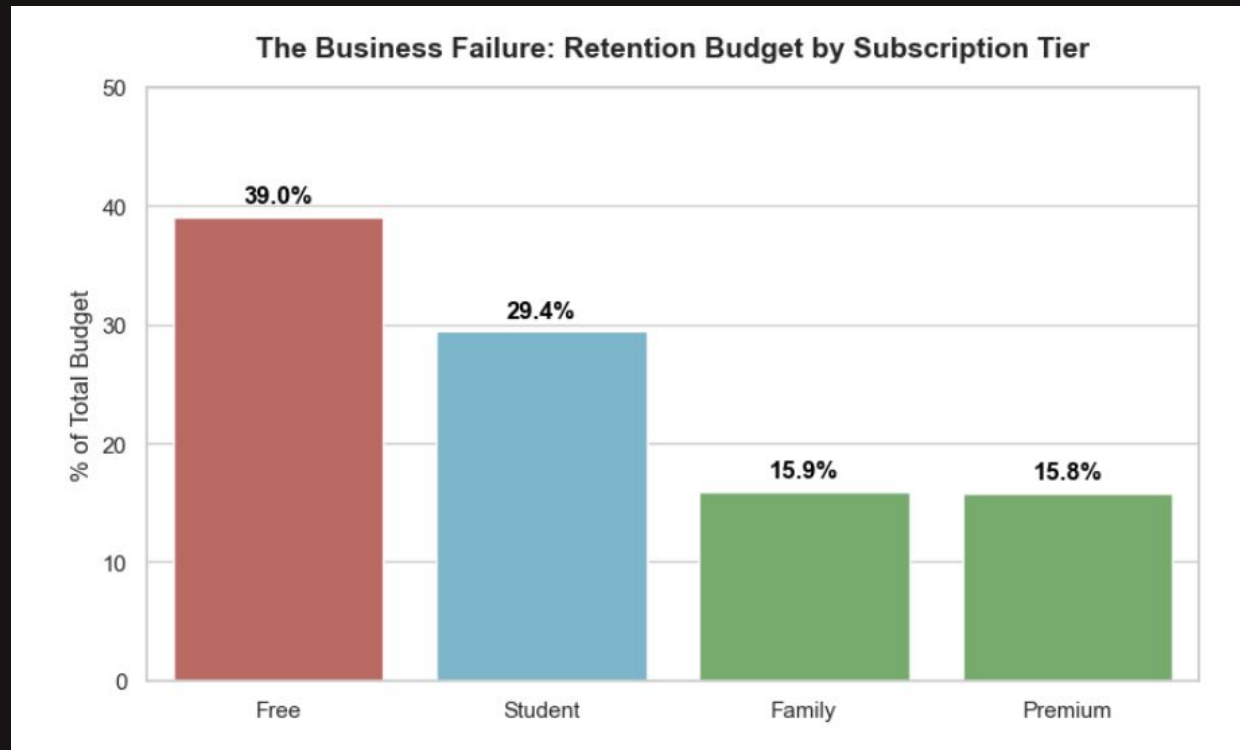


Table 6: Downstream Budget Allocation by Subscription Type

Subscription Type	Discounts Given	Budget (\$)	% of Total
Family	2,181	10,905	15.9%
Free	5,360	26,800	39.0%
Premium	2,166	10,830	15.8%
Student	4,040	20,200	29.4%

~40% of the retention budget is burned on non-paying users, while Premium revenue is neglected.

Verdict: Unfit for Deployment



Misaligned Incentives

Starves high-LTV customers while subsidizing Free users.



Proxy Discrimination

Usage hours and tiers function as proxies for age/income.

| Recommendations for a Responsible ADS

- ✓ **Data Collection:** Pivot from static snapshots to longitudinal "Usage Trajectories" to capture actual churn intent.
 - ✓ **Processing:** Implement threshold calibration per subgroup to equalize the False Negative Rate across tiers.
 - ✓ **Safeguards:** Introduce "Revenue-Aware Loss Functions" to prioritize retention for high-value customers.
 - ✓ **Governance:** Quarterly fairness re-audits to monitor demographic drift and proxy correlation.
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Thank you!